

Task Description

A Hospital Management System must generate a discharge bill for each type of patient (i.e. In-patient, out-patient). But depending on the patient, the billing rules change. For instance, for insurance billing is calculated based on insurance coverage rules, for a government health program the fixed subsidized rates are used, and the patient pays full if he/she pays on own. The system must be responsible for carrying out different billing policies on its own.

PlantUML

@startuml

```
abstract class Patient {  
    - name : String  
    - totalCost : double  
    - insuranceCoverage : double  
    - deductible : double  
    - programType : String  
    - corporateCode : String  
    + {abstract} Billing() : double  
}
```

```
class InPatitent extends Patient{  
    + Billing() : double  
}
```

```
class OutPatitent extends Patient{  
    + Billing() : double  
}
```

```
abstract class BillingStrategy {  
    + {abstract} calculateBill(patient : Patient) : double  
}
```

```

class InsuranceBilling extends BillingStrategy {
    + calculateBill(patient : Patient) : double
}

```

```

class SelfPayBilling extends BillingStrategy {
    + calculateBill(patient : Patient) : double
}

```

```

class GovernmentBilling extends BillingStrategy {
    + calculateBill(patient : Patient) : double
}

```

```

class BillingProcessor {
    - strategy : BillingStrategy
    + BillingProcessor(strategy : BillingStrategy)
    + generateBill(patient : Patient) : double
}

```

BillingProcessor "1" o-- "1" BillingStrategy : billingStrategy

Patient "0..*" --> "1" BillingProcessor

@enduml

Strategy Pattern Role	Hospital Scenario
Strategy Interface	BillingStrategy
Concrete Strategies	InsuranceBilling, SelfPayBilling, GovernmentBilling
Context class	BillingProcessor selects and uses the desired strategy
Patient	Chooses strategy based on patient information